PRE-LABORATORY LESSON · FIFTH-YEAR VETERINARY STUDENTS

Diagnostic Tools in Veterinary Epidemiology

Module 1 · Reference standards, test performance, prevalence, surveillance, and combined testing strategies



60–75 minutes · All numerical examples are hypothetical teaching data, not performance claims for particular commercial tests.

Diagnostic Tools in Veterinary Epidemiology

By the end of this lesson, you should be able to:



01

Evaluate reference standards

Explain their role and limitations.
High sensitivity and specificity alone do not make a test a gold standard.



02

Calculate test performance

Interpret sensitivity (Se), specificity (Sp), positive predictive value (PPV), and negative predictive value (NPV).



03

Interpret prevalence

With Se and Sp fixed: higher prevalence → higher PPV, lower NPV; lower prevalence → lower PPV, higher NPV.



04

Choose a testing strategy

Screening: prioritize sensitivity.
Confirmation: prioritize specificity.
Match tests to purpose and consequences of error.



05

Plan surveillance and timing

Explain monitoring, surveillance, and freedom from infection.
Consider sampling, test sensitivity, and testing intervals.



06

Combine diagnostic tests

Parallel (OR): increases sensitivity, reduces specificity.
Serial (AND): increases specificity, reduces sensitivity.



Interpret results in context: the animal, population, disease stage, and decision.

LESSON SEQUENCE

Where we're headed (75 minutes)



Carry the worked examples and checkpoint tables from this lecture into the computer laboratory — every calculation there has an expected answer here to check against.

SECTION 1

The Diagnostic Question & Reference Standards

Before choosing a test, decide what you want to detect, in which animals, and what you will do with the result.



START WITH THE DIAGNOSTIC QUESTION

Four different targets, four different tests

A test appropriate for one target may not answer another.

The pathogen

The pathogen itself or its genetic material.

Immune response

Evidence the animal was exposed — antibodies, cell-mediated response.

Clinical disease

Observable signs consistent with disease.

Herd infection

Whether infection is present anywhere within a group.



Veterinary example. A cow tests positive for antibodies against a pathogen. That result may indicate previous exposure or, depending on the assay and vaccine, vaccination. It does not automatically demonstrate that the cow is currently shedding the pathogen.

Ask students: *"If we want to identify animals currently shedding an organism, is an antibody test alone sufficient?"*

REFERENCE STANDARDS

What is a “gold standard”?

A reference standard is the best available method — or combination of methods — used to establish the target condition when evaluating another test. A test is not a gold standard simply because it is expensive, molecular, or reported to have high sensitivity and specificity.

Index test

The new rapid test for a bacterial infection in cattle.

Reference method

An established infection-status assessment that includes bacterial culture.

Limitation

Culture may miss infected animals when the specimen contains few viable organisms.

An animal that is rapid-test positive and culture negative might represent a rapid-test false positive — or a culture false negative.

> *“We evaluate a test against a reference standard, but we must also evaluate whether that reference standard is trustworthy.”*

SECTION 2

The 2×2 Table & Test Performance

One consistent worked example — 1,000 cattle, 100 infected — carries through sensitivity, specificity, PPV, and NPV.



BUILD THE DIAGNOSTIC 2×2 TABLE

1,000 cattle: 100 infected, 900 uninfected

	Infected	Uninfected	Total
Test positive	90 TRUE POSITIVE	45 FALSE POSITIVE	135
Test negative	10 FALSE NEGATIVE	855 TRUE NEGATIVE	865
Total	100	900	1,000

False negative

An infected animal is missed and may remain a source of infection.

False positive

An uninfected animal may undergo unnecessary isolation, investigation, or movement restrictions.

Which error is more serious?

It depends on the purpose and consequences of testing. Missing an infectious animal and wrongly restricting a valuable breeding animal involve different costs.

FOUR MEASURES, TWO STARTING POINTS

Sensitivity, specificity, PPV, and NPV

Measure	Question	Formula	Result
Sensitivity	Among infected animals, how many test positive?	TP ÷ (TP+FN)	**90%**
Specificity	Among uninfected animals, how many test negative?	TN ÷ (TN+FP)	**95%**
PPV	Among test-positive animals, how many are infected?	TP ÷ (TP+FP)	**66.7%**
NPV	Among test-negative animals, how many are uninfected?	TN ÷ (TN+FN)	**98.8%**

Se & Sp start from known status.
Use the infected / uninfected columns.

PPV & NPV start from the observed test result.
Use the positive / negative rows.

"Sensitivity: of 100 infected cattle, this test detects 90 and misses 10."

"PPV: of the cattle testing positive, approximately two-thirds are infected."

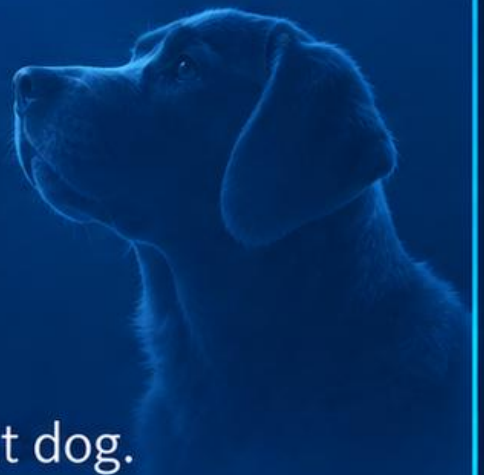
QUICK CHECK

A common misunderstanding

"90% sensitivity" ***does not*** mean that a positive animal has a 90% probability of infection. That question concerns **PPV**.

An owner asks:

"My dog tested positive. How likely is infection?"



Which measure answers the question?

Answer: PPV — interpreted using an appropriate pre-test probability for that dog.

SECTION 3

Prevalence & Predictive Values

Sensitivity and specificity are properties of the test. PPV and NPV also depend on how common infection is in the population tested.



FOR FIXED SENSITIVITY AND SPECIFICITY

Prevalence changes predictive values

HIGHER PREVALENCE

↑ higher PPV
↓ lower NPV

Where infection is likely, avoiding false negatives may be especially important.

LOWER PREVALENCE

↓ lower PPV
↑ higher NPV

Where infection is rare, avoiding false positives may be especially important.

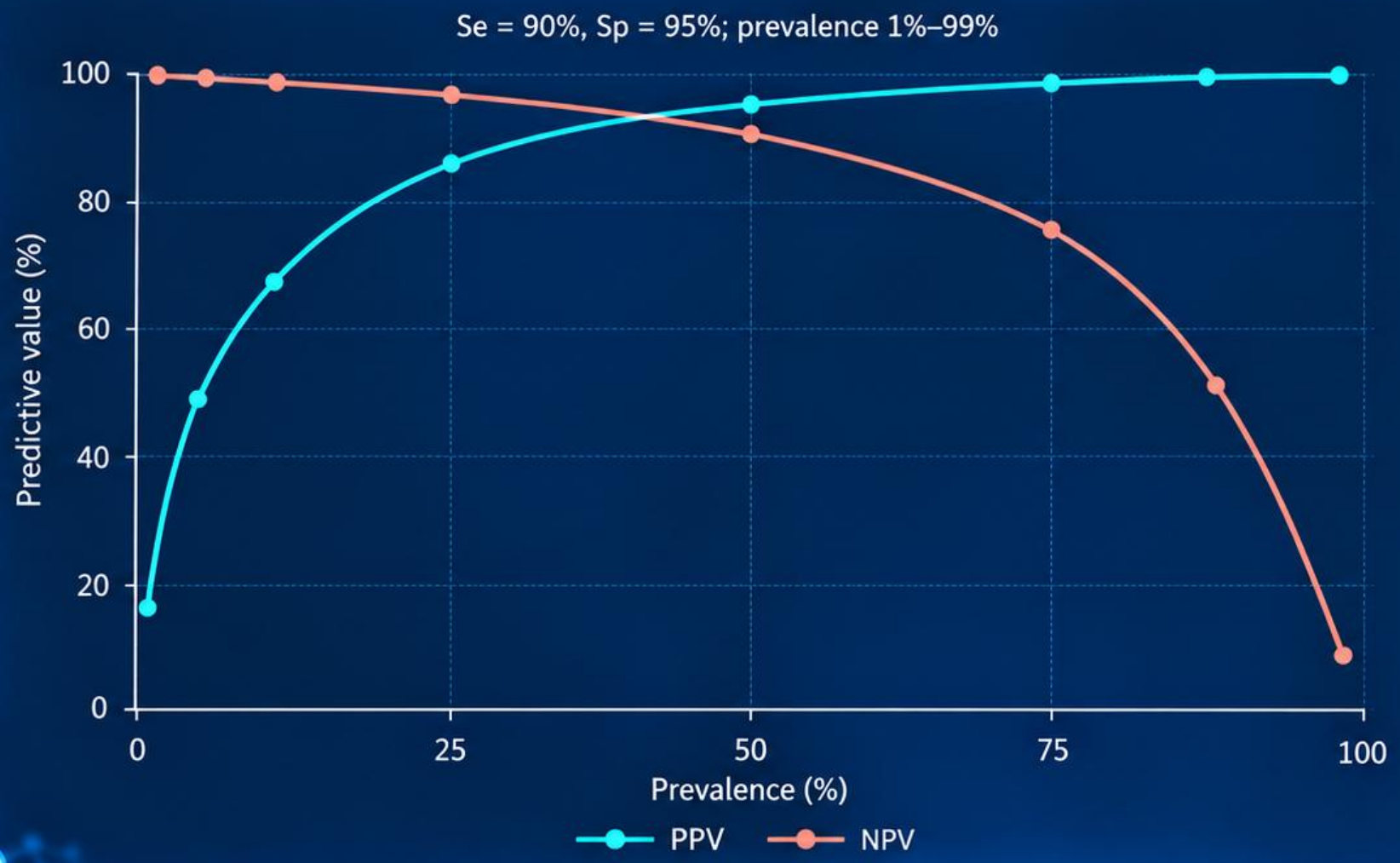
High prevalence does not itself produce high NPV, and low prevalence does not itself produce high PPV.

Worked example — Same test: Se 90%, Sp 95%. 10,000 cattle tested in each population.

Prevalence	TP	FN	FP	TN	PPV	NPV
1%	90	10	495	9,405	**15.4%**	**99.9%**
10%	900	100	450	8,550	**66.7%**	**98.8%**
50%	4,500	500	250	4,750	**94.7%**	**90.5%**

PLOTTING THE RELATIONSHIP

PPV rises, NPV falls, as prevalence increases



INTERPRETING THE CURVES

Example: 10,000 cattle tested.

At 1% prevalence:

9,900 uninfected cattle. A 5% false-positive rate yields 495 false positives, compared with 90 true positives.

At 50% prevalence:

500 infected cattle test negative. A negative result is less reassuring than at 1% prevalence.

DEFINITIONS & COMPUTER-LAB FORMULAS

Prevalence, incidence, and the PPV/NPV formulas

PREVALENCE

The proportion infected at a specified time or during a defined period.

INCIDENCE

New infections arising among animals at risk over time.

For predictive-value calculations, use prevalence or an appropriate individual pre-test probability. Do not substitute an incidence rate into the formula.

COMPUTER-LAB FORMULAS

Let P represent prevalence. Enter percentages as proportions (0.10 for 10%).

PPV

$$\frac{Se \times P}{Se \times P + (1-Sp) \times (1-P)}$$

NPV

$$\frac{Sp \times (1-P)}{(1-Se) \times P + Sp \times (1-P)}$$

SECTION 4

Selecting Tests, Surveillance & Timing

Purpose, specimen, disease stage, validation evidence, cost, turnaround time, and consequences of error all shape the choice.



WHICH DIAGNOSTIC TOOL SHOULD WE USE?

Match the test to the decision

Situation	Main concern	Teaching example
Screening	Missing infected animals	Screen incoming cattle with a sensitive approach, then investigate positives.
Confirmation	Wrongly classifying uninfected animals	Investigate a screening-positive breeding animal with a highly specific test.
High pre-test probability	A false-negative result	A clinically suspicious animal may need further work-up despite one negative.
Low pre-test probability	A false-positive result	A positive result in a low-risk herd requires careful confirmation.
Rapid outbreak investigation	Delayed detection and action	Weigh turnaround time alongside accuracy when choosing a field test.

Test performance should be established for the intended species and use — sensitivity and specificity are not immutable across populations and testing conditions.

CLASSROOM CHOICE EXERCISE

Two hypothetical tests: which for which job?

Test A

Sensitivity	**98%**
Specificity	**90%**

Test B

Sensitivity	**85%**
Specificity	**99.5%**

Question:

"Which would you initially consider for broad screening, and which for confirmation?"

Expected answer:

A for screening, B for confirmation — subject to validation, the testing algorithm, and consequences of errors.

"Choose the test for the decision you need to make."

Negative results alone don't prove zero infection

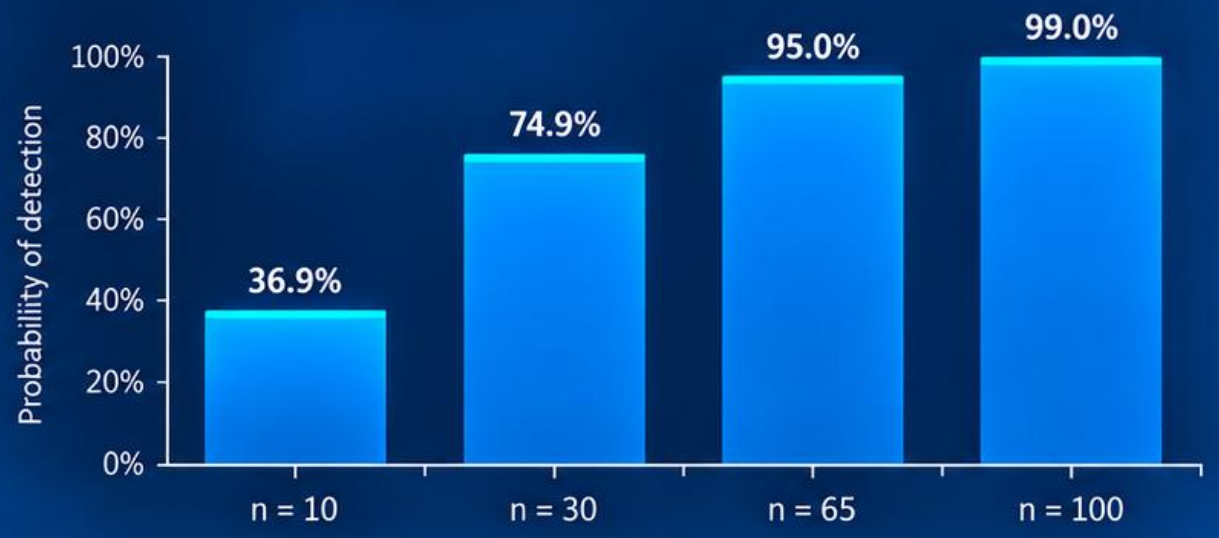
The strength of the evidence depends on sample size, animal selection, test performance, clustering, timing, and the infection level the survey is designed to detect.

WORKED EXAMPLE

Design prevalence 5%; individual-test sensitivity 90%.
Assuming independent, representative sampling from a large population.

P(infected and detected) = 0.05 × 0.90 = 0.045

P(detect ≥1 in n samples) = 1 − (1 − 0.045)^n



INTERPRETATION

Sampling only 10 animals
gives a substantial chance of missing infection despite a 90%-sensitive test (36.9% detection probability).

With 65 samples,
the design reaches approximately 95% detection probability if prevalence is really 5%.

This does not mean that negative results establish a 95% probability that the herd is completely free.

This calculation concerns detection of true infection. A practical surveillance system also needs to investigate false-positive results.

Repeated measurement vs. action-linked systems

Monitoring

Repeated measurement of health indicators.

Example: *a dairy farm records the percentage of cows with positive mastitis screening results each month.*

Surveillance

Collection, analysis, interpretation, and communication connected to decisions and action.

Example: *a regional programme collects abortion reports, analyses increases, and investigates when an alert threshold is reached.*

INTERPRETATION EXERCISE

Month	Positive	Tested	Proportion
January	10	100	**10%**
February	30	300	**10%**

"Did the proportion testing positive increase?"

No. The number of positive results increased because more animals were tested.

Then ask whether the months included comparable animals, specimens, tests, and reasons for testing.

TIME INTERVAL & TIMING OF TESTING

When was the animal exposed — and when would this test detect it?

Illustrative antibody-testing timeline



The day-3 result cannot confidently exclude recent infection. Follow-up timing should reflect the pathogen, assay, and appiicable protocol. The 14-day interval here is an illustrative assumption, not a universal retesting recommendation.

Surveillance interval example

For scheduled sampling every 30 days, an infection becoming detectable just after a visit could wait almost 30 days until the next visit, plus laboratory reporting time. Shorter intervals reduce detection delay, but frequency must match disease biology, risk, and resources.

SECTION 5

Parallel & Serial Testing

Defined by the rule for classifying results, not merely by whether the tests happen on the same day.



TWO COMBINATION RULES

Parallel (OR) vs. serial (AND)

PARALLEL · OR RULE

Positive if either test is positive.

↑ sensitivity, ↓ specificity.

Catches animals missed by one test but increases false-positive classifications — useful in a low-tolerance detection programme.

SERIAL · AND RULE

Positive only if both tests are positive.

↑ specificity, ↓ sensitivity.

Reduces false positives but may miss infected animals that fail either test — useful in low-prevalence screening.

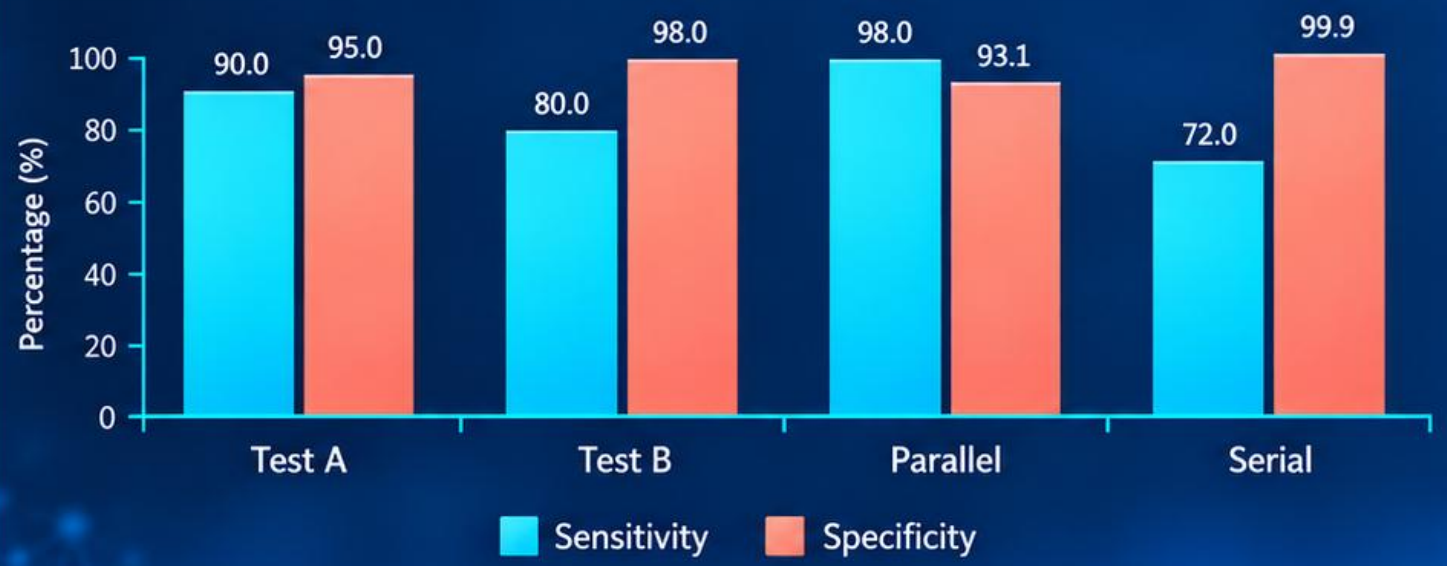
Test A	Test B	Parallel	Serial
Positive	Positive	Positive	Positive
Positive	Negative	Positive	Negative
Negative	Positive	Positive	Negative
Negative	Negative	Negative	Negative

A negative classification in the serial algorithm does not automatically resolve a clinically suspicious or discordant case.

WORKED CALCULATION

Combining Test A (Se 90 / Sp 95) & Test B (Se 80 / Sp 98)

Strategy	Sensitivity formula	Specificity formula	Result (Se / Sp)
Parallel	1 – (1–SeA) × (1–SeB)	SpA × SpB	**98.0% / 93.1%**
Serial	SeA × SeB	1 – (1–SpA) × (1–SpB)	**72.0% / 99.9%**



IMPORTANT QUALIFICATION

The multiplication formulas rely on conditional independence. Tests measuring similar biological responses — or repeating the same assay on the same specimen — may have correlated errors. Their actual combined performance must be evaluated accordingly.

Demonstrated in veterinary diagnostic research on parallel/serial testing for canine Leishmania infection.

SECTION 6

Before You Start the Computer Lab

Five readiness questions, then six activities to reproduce in Excel, R, or your software of choice.



READINESS CHECK

Five questions before students start

1	Does 95% sensitivity mean a test-positive animal has a 95% probability of infection?	**A:** *No — that is a predictive-value question.*
2	What happens to PPV when prevalence decreases, Se and Sp unchanged?	**A:** *PPV decreases.*
3	Which strategy uses an OR rule?	**A:** *Parallel testing.*
4	Do negative results from 10 sampled cattle prove herd freedom?	**A:** *No — sampling and detection probability must be considered.*
5	Can a negative result shortly after exposure exclude infection?	**A:** *Not necessarily — the target may not yet be detectable.*

COMPUTER LABORATORY ACTIVITIES

Six activities — submit an interpretation with every number

#	Activity	Expected result
1	Calculate test performance (TP 90, FP 45, FN 10, TN 855)	Se 90%; Sp 95%; PPV 66.7%; NPV 98.8%
2	Change prevalence — recompute PPV/NPV at 1%, 10%, 50%	Reproduce the checkpoint table (Section 3)
3	Plot PPV and NPV against prevalence, 1%–99%	PPV rises; NPV falls
4	Combine Test A and Test B: parallel and serial performance	Parallel 98.0%/93.1%; serial 72.0%/99.9%
5	Detection probability at n = 10, 30, 65 (5% prevalence, 90% Se)	≈ 36.9%, 74.9%, 95.0%
6	Recommend & justify a screening strategy in three sentences	Address missed infections, false positives, follow-up

BEFORE YOU RUN ANY CALCULATION IN LAB

For every result, state the denominator, explain the biological meaning, and identify the decision it supports.

FURTHER READING CITED IN THIS LECTURE

- WOAH — diagnostic assay validation guidance
- FDA — statistical guidance for diagnostic test evaluation
- Merck Veterinary Manual — calculating Se/Sp
- WOAH — animal health surveillance guidance
- Parallel/serial testing in canine Leishmania infection (peer-reviewed)

All numerical examples in this lecture are hypothetical teaching data, not performance claims for particular commercial tests.